

Physics, 2018 EC, entrance exam, version 1

Grade 12 · Physics · 2018 E.C.

Time allowed: 120 minutes

Questions: 46

Total marks: 46

Name _____ Grade _____ ID No. _____ Date _____

1. Which of the following is not true about uniform motion? (1)
 - A. It covers equal interval of distance at equal interval of time in a fixed direction.
 - B. Its speed remains constant throughout its motion along straight line.
 - C. Its acceleration stays uniform throughout its motion on its way at a given time.
 - D. Its graph is a straight line with constant positive slope.
2. Two students walk from school to home. Student A takes a short cut in straight line, but student B follows the road which curves around a hill. Which one is true? (1)
 - A. Student A has greater distance but the same displacement
 - B. Student B has greater distance but the same displacement
 - C. Student A has greater displacement but the same distance.
 - D. Both distance and displacement are the same for both students.
3. Which statement is wrong about average velocity and instantaneous velocity? (1)
 - A. Average velocity depends on total displacement, instantaneous velocity does not.
 - B. Average velocity and instantaneous velocity are equal when an object moves in straight line with constant velocity.
 - C. The speed shown on a speedometer at a moment represents instantaneous velocity.
 - D. The average velocity expresses motion of an object at specific moment in time.
4. If the force of attraction between two point masses at a distance r is 250N, what will be the new force if the distance between them decreases by quarter for the same masses? ($G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$) (1)

A. 1600N	B. 8000N
C. 4000N	D. 1200N
5. All are dimensionless except (1)

A. Specific gravity	B. Young's modulus
C. Strain	D. Index of refraction
6. Which of the following is necessarily true? (1)
 - A. If the velocity of an object is zero, then acceleration is also zero.
 - B. If acceleration of an object is zero then its velocity is also zero.
 - C. Deceleration means that acceleration is necessarily negative.
 - D. The instantaneous acceleration has the same value as the average acceleration if the motion is uniformly accelerated motion.
7. Which of the following sets of quantities remains constant in uniform circular motion? (1)

A. Velocity, speed, radius	B. Velocity, acceleration, kinetic energy
C. Speed, acceleration, radius	D. Speed, radius, kinetic energy

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8. An earth satellite has an orbit radius which is 4 times that of a communication satellite C. What is the period of revolution of the earth's satellite in terms of communication satellite C? (use period of the earth = 1 year) (1)
- A. $T_{es} = 8T_c$ B. $T_{es} = 4T_c$
C. $T_c = 5T_{es}$ D. $T_c = 2T_{es}$
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9. The acceleration that a force produces on 80kg mass is larger by 6 m/s^2 than the acceleration it produces on 100kg. What will be the acceleration on 80kg mass? (1)
- A. 24 m/s^2 B. 18 m/s^2
C. 30 m/s^2 D. 40 m/s^2
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10. When a driver applies a sudden brake to save a child at the center of the road, you are moving forward from your seat. Which Newton's law expresses this phenomenon best? (1)
- A. Newton's first law B. Newton's second law
C. Newton's third law D. Newton's gravitational law
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11. How much work is required to accelerate a 900kg car from 10 m/s to 20 m/s ? (Ignore friction) (1)
- A. $7.46 \times 10^5 \text{ J}$ B. $2.46 \times 10^5 \text{ J}$
C. $1.35 \times 10^5 \text{ J}$ D. $4.36 \times 10^5 \text{ J}$
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12. A horizontal force of 80N is applied on a 10kg block which is moving on a horizontal rough surface. If the coefficient of friction between the block and the surface is $\mu = 0.5$, what is the acceleration of the block? (1)
- A. 8 m/s^2 B. 2 m/s^2
C. 18 m/s^2 D. 3 m/s^2
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13. Which one of the following is false? (1)
- A. The slope of acceleration versus time graph is velocity
B. The area under velocity versus time graph is displacement
C. The area under acceleration versus time graph is velocity
D. The slope of velocity versus time graph is acceleration
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14. In which one of the following cases is work done in physics? (1)
- A. Holding a 200N load without moving.
B. Pushing a concrete wall horizontally by applying horizontal force.
C. Pushing a block horizontally by applying horizontal force.
D. Carrying a load by shoulder and moving along straight line a distance of 2km.
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15. Which one of the following correctly expresses friction strength? (1)
- A. Sliding friction > rolling friction > static friction
B. Sliding friction > rolling friction
C. Sliding friction > friction > static friction
D. Rolling friction < sliding friction < static friction
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16. An object of mass is dropped from a certain height to the ground; at what height above the ground do we assume that the object has the same potential and kinetic energies if its velocity is 20 m/s ? (1)
- A. 5 m B. 20 m
C. 60 m D. 40 m

17. Which one of the following is NOT true about the three vectors? (1)

- A. $-A = B + C$
C. $A + B - C = 0$
- B. $B = -C - A$
D. $A + B + C = 0$

18. What will be the magnitude of the resultant and direction of the given vectors below? 5N at 53° and 10N at 37° (1)
($\sin 37^\circ = \cos 53^\circ = 0.6$, $\cos 37^\circ = \sin 53^\circ = 0.8$)

- A. $2\sqrt{5}$ and $\tan^{-1}2$, N of E
B. $5\sqrt{5}$ and $\tan^{-1}2$, N of E
C. 5 and $\tan^{-1}2$, N of E
D. $15N$ and $\tan^{-1}(1/2)$, E of N

19. A car initially at rest is uniformly accelerated along a straight road. The car passes two marks separated by 48m at 4s and 8s respectively. What is the acceleration of the car? (1)

- A. 1 m/s^2 B. 3 m/s^2
C. 2 m/s^2 D. 4 m/s^2

20. An automobile driver spotted a truck 250m ahead of him moving on a straight level road. If the truck is moving uniformly at a velocity of 10 m/s, how long will it take the automobile driver to overtake the truck if he drives at 20 m/s?

- A. 10 s
B. 50 s
C. 5 s
D. 25 s

21. Which of the following is False? (1)

- A. Materials with higher Young's modulus also possess greater bulk modulus.
- B. Materials with higher strain are highly elastic.
- C. Shearing stress happens when the deforming forces are tangential.
- D. Materials with smaller strain are highly elastic.

22. A 400kg load is hung on a wire having a length of 6.0m and cross-sectional area of $0.40 \times 10^{-5} \text{ m}^2$ and Young's modulus $8.00 \times 10^{10} \text{ N/m}^2$. What is its increase in length? (1)

- A. 0.075 m
B. 7.25 m
C. 0.09 m
D. 0.05 m

23. Two charges of $+9\mu\text{C}$ and $+25\mu\text{C}$ are 40cm apart. At what distance from either charge is the field zero? (1)

- A. 15 cm from q_1 and 25 cm from q_2
- B. 10 cm from q_1 and 15 cm from q_2
- C. 35 cm from q_1 and 15 cm from q_2
- D. 20 cm from q_1 and 45 cm from q_2

24. Suppose a given resistor is connected to a constant voltage supply. When will the current in the resistor decrease? (1)

- A. When the length of the resistance is reduced.
B. When the resistance of the resistor is decreased.
C. When cross-sectional area of the resistor increased.
D. When the length of the resistor is increased.

25. From the given circuit below, the current drop in 6Ω and 12Ω resistors respectively is (1)

- A. 3A and 2A
B. 4A and 3A
C. 1A and 0.5A
D. 1.2A and 3A

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26. A wheel starts rotating at 1.2 Hz to 1.8 Hz in 2 seconds. What will be the angular displacement it covers within 2 seconds? (1)
- A. 3 rad
B. 6 rad
C. 4 rad
D. 0.3 rad
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27. An egg is placed at the bottom of a vessel filled with water. Common salt is slowly added and dissolved in the water until the egg rises and floats at the surface. From this information, what can you conclude? (1)
- A. The density of salt exceeds the density of the egg.
B. There is a tendency for calcium in the eggshell to be repelled by sodium chloride.
C. Salt sinks to the bottom.
D. There is no physical explanation.
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28. A substance whose density is equal to $0.4 \times 10^3 \text{ kg/m}^3$ floats in fresh water ($\rho_w = 1000 \text{ kg/m}^3$). What fraction of its volume is submerged? (1)
- A. $1/2$
B. $1/5$
C. $2/5$
D. $1/4$
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29. In the figure, the time it takes for the planet to go from A to B, C to D, and E to F is the same. Compare the areas A_1 , A_2 , and A_3 in terms of size. (1)
- A. $A_1 \neq A_2 \neq A_3$
B. $A_1 = A_2 = A_3$
C. $A_1 = A_2 > A_3$
D. $A_1 > A_2 = A_3$
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30. Physics can be applied to everyday life except? (1)
- A. The development of telephone, telegraph and telex have enabled us to transmit important messages instantly.
B. The development of radio, television, satellites have increased the means of communication.
C. Rapid means of transport have increased the pace of transportation through air, water, and land.
D. Enhancing artistic expressions and career opportunities by ignoring real world implications.
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31. Which one is wrong action when a leak is suspected in high-pressure equipment? (1)
- A. Shut down and depressurize the system safely.
B. Use hands to feel for escaping fluid from the system.
C. Keep a safe distance from the suspected leak.
D. Report the issue according to safety procedures.
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32. Which of the following best describes the importance of career awareness in physics education? (1)
- A. It is optional and has little impact on future employment.
B. It is irrelevant; students should only focus their theoretical knowledge.
C. It helps students understand the diverse career opportunities, provide hands-on experience and networking system.
D. It primarily focuses students' interest in academia and it serves no purpose in advancing physics knowledge.
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33. If you feel a pressure three times that of the atmospheric pressure when you are deep under water, how high is the water level from your position? ($\text{atm} = 10^5 \text{ Pa}$, density $\rho_w = 10^3 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$, and assume atmospheric pressure is exerted on it.) (1)
- A. 20 m
B. 10 m
C. 40 m
D. 30 m

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- 34.** The vertical component of force acting at an angle of 37° to the horizontal is 12N. What will be the magnitude of the force and its horizontal component respectively? (1)
- A. 50N, 30N
B. 20N, 16N
C. 20N, 25N
D. 5N, 25N
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- 35.** For what value of C lying on the x-axis does $A \cdot (B+C) = 0$ if $A = 5i - 3j + 4k$ and $B = 2i + 4j - 5k$? (1)
- A. $5i$
B. $-3.2i$
C. $1.9i$
D. $-4.4i$
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- 36.** Which of the following is NOT correct about momentum? (1)
- A. The total linear momentum of an isolated system remains constant.
B. Impulse is the change in momentum.
C. Impulse and momentum have the same unit.
D. Linear momentum is mass multiplied by impulse.
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- 37.** In aerospace engineering, the stability of a satellite in orbit is ensured by (1)
- A. Air friction
B. Newton's law of motion and gravitational force
C. Electromagnetic shielding
D. Laser production
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- 38.** What is the main focus of physics compared to chemistry? (1)
- A. Physics studies reactions between substances but chemistry studies motion.
B. Physics studies fundamental laws of matter and energy, while chemistry studies composition and reaction of matter.
C. Physics studies at macro levels about atoms while chemistry studies at micro levels.
D. Physics concerns transformation of physical matter but chemistry concerns the movement of physical objects.
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- 39.** The principle behind optical fibers in medical imaging (endoscopy) is (1)
- A. Total internal reflection
B. Diffraction
C. Polarization
D. Scattering
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- 40.** Which of the following best describes cruise missiles? (1)
- A. They are propelled by rocket.
B. It follows a high space-based trajectory before descending to target.
C. It travels at a very high speed without guidance.
D. It flies at low altitudes with continuous engine power and precise guidance.
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- 41.** A body is thrown vertically upward and at the instant when it reaches its maximum height; (1)
- A. Its velocity is zero and its acceleration is constant
B. Its velocity is zero and its acceleration is also zero.
C. Its velocity is zero but its acceleration is maximum.
D. Its velocity is zero but its acceleration is minimum.
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- 42.** A particle moves in a circular path of radius 0.1m with a constant angular speed of 10π rad/s. What is the acceleration of the particle? (1)
- A. $10\pi^2$ m/s²
B. $20\pi^2$ m/s²
C. $30\pi^2$ m/s²
D. $40\pi^2$ m/s²

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- 43.** Three balls A, B and C are thrown at an angle 37° , 45° and 53° respectively from the ground with the same initial velocity u . Which of the following comparisons about the horizontal distance they travel is correct? (1)
- A. Distance of A = Distance of B < Distance of C
 - B. Distance of A = Distance of C < Distance of B
 - C. Distance of A = Distance of C > Distance of B
 - D. Distance of B < Distance of C < Distance of A
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- 44.** Which one of the following is false about fluid flow? (1)
- A. Laminar fluid flow occurs with low velocity and small diameter pipe.
 - B. Turbulent fluid flow occurs with high velocity and larger diameter pipe.
 - C. The path of different particles during laminar fluid flow crosses each other.
 - D. The velocity of the particles is tangent to streamline for laminar fluid flow.
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- 45.** The V-t graph of a moving object as shown in the figure below. The maximum acceleration is (1)
- A. 1 cm/sec^2
 - B. 2 cm/sec^2
 - C. 3 cm/sec^2
 - D. 6 cm/sec^2
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- 46.** A solid sphere of radius R and moment of inertia about its diameter ($I = \frac{2}{5} mr^2$) starts from rest and rolls down an inclined plane of 7m high. What is the speed of the sphere at the bottom of the inclined plane? (Ignore friction) (1)
- A. 10 m/s
 - B. 100 m/s
 - C. 30 m/s
 - D. 5 m/s

— END OF EXAMINATION —